

# descriptive analysis of solar panel data

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## 1 introduction

Since 2022, my parents have had solar panels installed on their roof. This report provides a descriptive analysis of the solar panel data and examines several practical questions related to electricity production, household consumption, grid usage, battery use, and seasonal variation.

A common assumption is that owning a solar panel system automatically leads to substantial financial benefits. However, the actual value of such a system depends on how much electricity is produced, how much of it is used directly, how much is stored in the battery, and how much still has to be imported from the public grid. This analysis therefore takes a data-based look at the system and evaluates how efficiently the generated solar energy is used.

The main goal is to better understand the cost-efficiency of the system. At the same time, this can also be interpreted as a measure of ecological efficiency: the more locally generated solar energy is used by the household, the less dependent the household is on externally supplied electricity.

A second motivation is to assess the current degree of grid dependence. In particular, the analysis investigates how much of the household's electricity demand can be covered by solar production and battery storage, and how much still depends on electricity imported from the public grid.

### 1.1 structure of the data

Each row represents one aggregated 3-hour measurement interval. The timestamp marks the beginning of the interval. Based on the hour of the timestamp, each row is assigned to a daytime category.

```

dat$daytime <- NA

dat[dat$hour %in% c(21, 0, 3), ]$daytime <- "night"
dat[dat$hour == 6, ]$daytime <- "morning"
dat[dat$hour == 9, ]$daytime <- "late_morning"
dat[dat$hour == 12, ]$daytime <- "midday"
dat[dat$hour == 15, ]$daytime <- "afternoon"
dat[dat$hour == 18, ]$daytime <- "evening"

```

The most important variables are:

- production [Wh]: energy generated by the solar panels.
- consumption [Wh]: energy consumed by the household.
- battery\_charge [Wh]: energy used to charge the 10 kWh battery.
- battery\_discharge [Wh]: energy supplied by the battery to the household.
- grid\_feedin [Wh]: energy exported to the public grid.
- grid\_consumption [Wh]: energy imported from the public grid.
- battery\_state\_of\_charge [%]: current battery charge level.
- direct\_consumption [Wh]: solar energy consumed directly by the household.

## 1.2 data cleaning

The raw dataset does not contain the same number of observations for every year-month combination. This is expected in part, because months have different numbers of days. A complete day contains eight observations, so January should contain  $31 * 8 = 248$  observations, whereas February contains either  $28 * 8 = 224$  observations or, in a leap year,  $29 * 8 = 232$  observations.

The following table shows the number of observations per year and month before cleaning:

```

table(dat$year, dat$month)

##
##      1  2  3  4  5  6  7  8  9 10 11 12
## 2023  0  0 165 240 248 240 248 248 240 248 240 248
## 2024 248 232 248 240 248 240 248 226 240 248 240 248
## 2025 248 224 248 240 248 240 248 248 240 248 240 248
## 2026 248 221 248 240 208  0  0  0  0  0  0  0

```

The table shows that some months are incomplete. March 2023 is incomplete because data collection started during that month, so it is removed from the analysis.

```

dat <- dat[!(dat$year == 2023 & dat$month == 3), ]

```

Other months contain only a small number of missing 3-hour intervals. For these months, the missing values are approximated using the mean from the same month, day, and hour in other years. In total, 65 out of 9256 are approximated.

```

table(dat$year, dat$month)

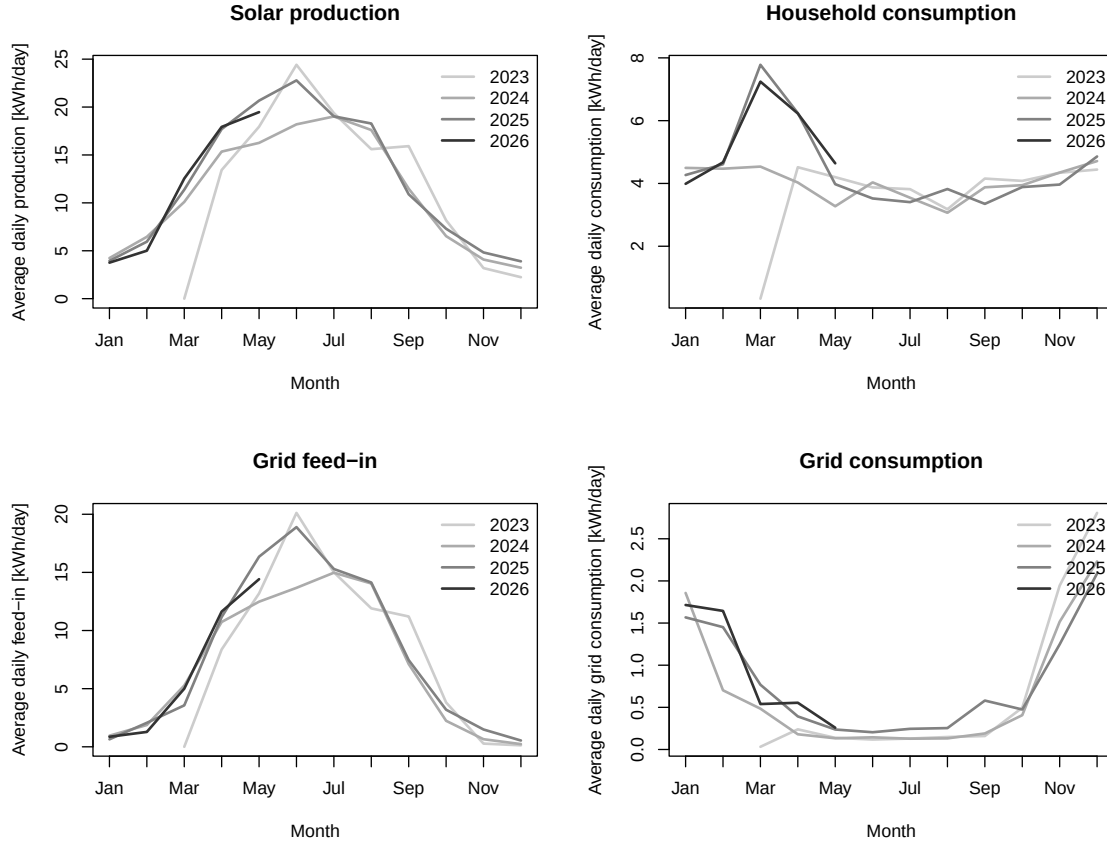
##
##      1  2  3  4  5  6  7  8  9 10 11 12
## 2023  0  0  0 240 248 240 248 248 240 248 240 248
## 2024 248 232 248 240 248 240 248 248 240 248 240 248
## 2025 248 224 248 240 248 240 248 248 240 248 240 248
## 2026 248 224 248 240 248  0  0  0  0  0  0  0

```

The resulting dataset preserves the correct calendar structure of the data while removing distortions caused by incomplete months or missing 3-hour intervals. This makes seasonal comparisons of absolute production and consumption values more meaningful.

## 2 descriptive analysis

First, the data are inspected from a yearly perspective to identify whether seasonal categories can be derived from the data. Because months have different numbers of days, the monthly plots use average daily values instead of simple monthly sums.



The plots show strong seasonality in the energy flows. Solar production and grid feed-in peak in the brighter months and decline sharply in winter. As a result, grid consumption remains very low during spring and summer, when solar availability is high, and rises again in autumn and winter as solar production declines.

Grid consumption and solar production appear to move in opposite directions, while household consumption remains comparatively stable over the year. One notable exception is the pronounced spike in household consumption in the 2025 and 2026 data. This pattern is addressed in the financial section of the report.

### 2.1 defining solar seasons

Instead of using fixed calendar seasons, seasons are defined directly from the data as high-production and low-production solar months.

- **Warm months:** the six months with the highest average monthly solar production.
- **Cold months:** the six months with the lowest average monthly solar production.

The average monthly values are calculated in two steps. First, production and grid consumption are summed within each year-month combination. Second, these monthly totals are averaged across years. This avoids giving more weight to years with more observations.

The table ranks months by average monthly solar production. Months with ranks 1 to 6 are classified as warm months, while the remaining months are classified as cold months. To evaluate whether high-production months also tend to have lower grid consumption, I calculated the Spearman correlation between the production ranking and the grid-consumption

Table 1: Estimated average monthly solar production and grid consumption, ranked by average monthly production

| month | production | grid_consumption | n_years_observed | month_name | production_rank | grid_consumption_rank | g |
|-------|------------|------------------|------------------|------------|-----------------|-----------------------|---|
| 6     | 653877     | 4671             | 3                | June       | 1               | 12                    | v |
| 7     | 593285     | 5200             | 3                | July       | 2               | 11                    | v |
| 5     | 576202     | 5954             | 4                | May        | 3               | 9                     | v |
| 8     | 532073     | 5513             | 3                | August     | 4               | 10                    | v |
| 4     | 483020     | 10257            | 4                | April      | 5               | 7                     | v |
| 9     | 382921     | 9303             | 3                | September  | 6               | 8                     | v |
| 3     | 351404     | 18617            | 3                | March      | 7               | 5                     | c |
| 10    | 227380     | 14135            | 3                | October    | 8               | 6                     | c |
| 2     | 164636     | 35566            | 3                | February   | 9               | 4                     | c |
| 1     | 123465     | 53761            | 3                | January    | 10              | 2                     | c |
| 11    | 121013     | 46908            | 3                | November   | 11              | 3                     | c |
| 12    | 96810      | 73243            | 3                | December   | 12              | 1                     | c |

Table 2: Total energy flows over the full observation period.

| Energy flow        | MWh   |
|--------------------|-------|
| Solar production   | 13.98 |
| Total consumption  | 5.03  |
| Battery charge     | 2.57  |
| Battery discharge  | 1.80  |
| Grid feed-in       | 9.05  |
| Grid consumption   | 0.87  |
| Direct consumption | 2.50  |

ranking. The resulting correlation is -0.97, which indicates a strong inverse seasonal relationship between solar production and grid dependence.

Based on this classification, we can enrich the data set by adding season as a new variable.

```
dat$season <- ifelse(dat$month %in% 4:9, "warm", "cold")
dat$season <- factor(dat$season)
```

## 3 household energy balance

This section summarizes the household energy balance. First, it presents total energy flows over the full observation period. Then, it shows how these flows differ between warm and cold solar seasons.

### 3.1 total energy balance

The total values provide an overview of the full household energy balance. However, they do not show how much of the total is driven by warm or cold months. For that reason, the total table is followed by a seasonal breakdown.

### 3.2 seasonal breakdown of household energy flows

The total values hide important seasonal differences. Solar production is expected to be higher during warm months, while grid consumption is usually more relevant during cold months.

To compare seasonal energy flows without bias from different observation lengths, the analysis first aggregates all 3-hour observations into monthly totals. These monthly totals are then grouped into warm and cold seasons for each year.

Incomplete seasons are standardized to a six-month equivalent before the seasonal totals are averaged across years. The flowcharts therefore show the average total energy flows of a typical warm or cold season, not average monthly values and not simple totals over the observed period.

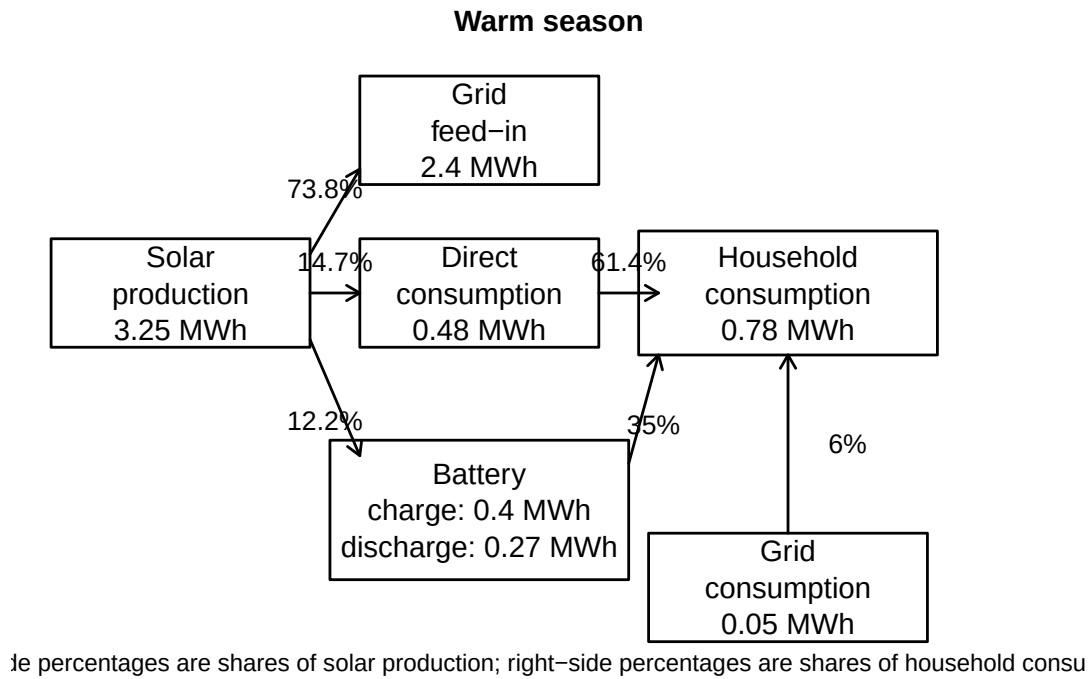


Figure 1: Average six-month energy flows in the warm season.

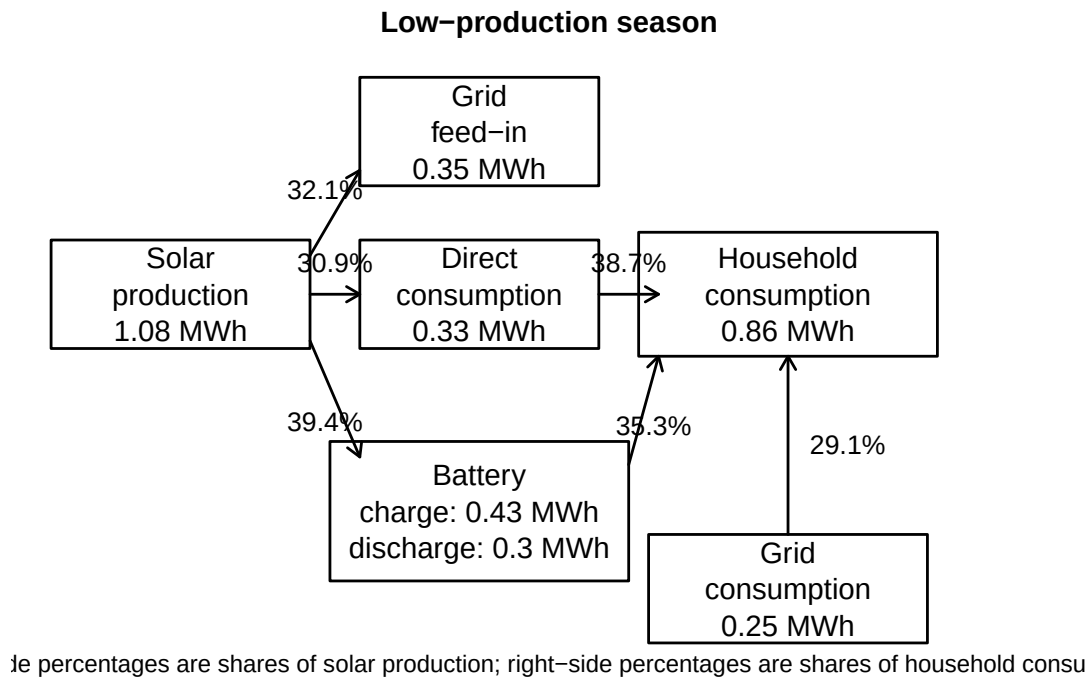


Figure 2: Average six-month energy flows in the low-production season.

Table 3: Estimated grid-related financial balance for an average six-month warm or cold season.

| Season | Grid feed-in [MWh] | Grid consumption [MWh] | Feed-in revenue [EUR] | Grid cost [EUR] | Balance [EUR] |
|--------|--------------------|------------------------|-----------------------|-----------------|---------------|
| cold   | 1.08               | 0.25                   | 17.34                 | 87.81           | -70.46        |
| warm   | 3.25               | 0.05                   | 119.97                | 16.47           | 103.50        |

### 3.3 financial implications of seasonality

The financial implications of grid exchange are asymmetric because electricity bought from the grid is much more expensive than electricity sold to the grid. Assuming a feed-in compensation of 0.05 EUR/kWh and a grid electricity price of 0.35 EUR/kWh, one kilowatt-hour imported from the grid costs seven times as much as one kilowatt-hour exported to the grid earns.

This shows why direct consumption and battery use are financially more valuable than feeding electricity into the grid. Each kilowatt-hour used directly by the household avoids buying electricity for 0.35 EUR, whereas feeding the same kilowatt-hour into the grid would only earn 0.05 EUR. Therefore, using solar electricity locally is worth about 0.30 EUR more per kWh than exporting it. With this background, we can calculate the costs we lose by feeding electricity into the grid instead of using it ourselves.

```
# total opportunity cost per exported kWh [in EUR]
wh_to_kwh(sum(dat$grid_feedin, na.rm = TRUE)) * 0.30
```

```
## [1] 2713.754
```

```
# in the warm season [in EUR]
wh_to_kwh(
  sum(dat[dat$season == "warm", "grid_feedin"], na.rm = TRUE)
) * 0.30
```

```
## [1] 2401.842
```

```
# in the cold season [in EUR]
wh_to_kwh(
  sum(dat[dat$season == "cold", "grid_feedin"], na.rm = TRUE)
) * 0.30
```

```
## [1] 311.9121
```

This opportunity cost was the main reason why my parents bought a “Heizstab” to heat water with excess solar electricity. As we already know the excess electricity makes up almost three quarters of production in summer and about one third in winter.

This also puts the recent political debate about reducing support for solar feed-in into perspective. The data show that solar panels are not simply an easy way to make money. Without local self-consumption, feeding electricity into the grid can be economically unattractive because the household gives up electricity that would otherwise replace much more expensive grid electricity.

## 4 improving daily energy usage efficiency